Exercise: Determining the Age, Metallicity and Distance of a Mystery Globular Cluster

March 5th, 2026

Objective

In this assignment, you will determine the age, metallicity, and distance to an unidentified globular cluster that is located in the Milky Way. You will do this by visually fitting theoretical stellar evolution models (isochrones) to the cluster's observational Color-Magnitude Diagram (CMD).

The Data

In the class github repo, you will find a photometry file named `mystery_gc.dat`. The data was taken using the Hubble Space Telescope's (HST) Advanced Camera for Surveys (ACS).

- **Column 1:** F606W magnitude (roughly equivalent to broad R-band)
- **Column 2:** F814W magnitude (roughly equivalent to I-band)
- **Column 3 & 4:** Photometric errors in the F606W and F814W bands

A CMD and Hess diagram of the mystery cluster are shown in Figure 1.

Instructions

Part 1: Generating Theoretical Isochrones

We will use the PARSEC (PAdova and TRieste Stellar Evolution Code) isochrone database.

1. Use a python API (e.g., `ezpadova`) or go to the CMD web interface: `http://stev.oapd.inaf.it/cgi-bin/cmd`
2. For **Photometric system**, select *HST/ACS WFC*.

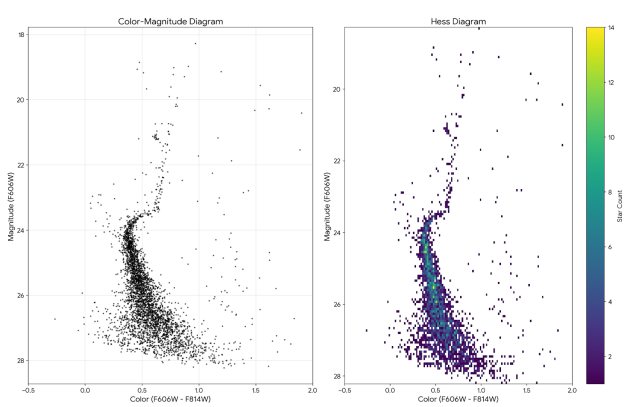

Figure 1: **Left:** A CMD of the mystery globular cluster in HST's ACS F606W and F814W bands. **Right:** A Hess diagram, i.e., a 2D binned CMD, of the mystery cluster.

3. Under **Isochrones**, you will need to test different ages and metallicities. Recall, that globular clusters are generally old ($\gtrsim 5$ Gyr) and metal-poor compared to the Sun. The solar metallicity is approximately $Z_\odot \approx 0.0152$ or [Fe/H]$= 0$.
4. Generate and download a grid of isochrones over a reasonable range of ages and metallicites. Your knowledge of stellar evolution and the CMD in Figure 1 may help guide your choices.

Part 2: Visual Fitting

Write a script and/or use your AI assistant to overlay a set theoretical isochrones onto a CMD of the mystery cluster in order to estimate it's age, metallicity, and distance.

The PARSEC models provide absolute magnitudes (M_{F606W} and M_{F814W}). However, your data is in apparent magnitude (m_{F606W} and m_{F814W}). To align the theoretical isochrone with the observed data, you must apply a **Distance Modulus** (μ) shift to the y-axis, and a **Reddening** shift to both axes.

For this assignment, assume reddening is $A_{F606W} = 0.06$ mag and $A_{F814W} = 0.04$ mag. The distance modulus is $\mu = m - M = 5\log_{10}(d) - 5$, where d is in units of parsecs.

Part 3: Parameter Estimation

Adjust your isochrone parameters (Age, Z or [Fe/H], and Distance Modulus) until you find the best visual fit. Pay special attention to matching these critical features:

- The Main Sequence Turn-Off (MSTO) point.
- The slope and curvature of the Red Giant Branch (RGB).
- The Subgiant Branch (SGB).

Findings

Place one slides in the class slide with a plot showing the cluster CMD and with an isochrone overplotted for you best estimated age, metallicity, and distance.